

DONESH DAVATGAR

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Projects

From The Ashes: Reborn | Unreal Engine C++, Gameplay Ability System

- Engineered a combat system in C++ using Unreal Engine's Gameplay Ability System (GAS), implementing combo chaining, mid-combo branching, aerial combat, and multi-weapon functionality inspired by Devil May Cry and Metal Gear Rising.
- Developed a modular, weapon-centric character architecture where each equipped weapon dynamically defines its ability sets, animation layers, gameplay tags, and attribute configurations, enabling weapon swapping and data-driven customization.
- Created an Enemy AI system inspired by DOOM-style token system with C++ that regulates enemy attack permissions, ensuring staggered, coordinated assaults for intelligent but fair interactions that enables a one-vs-many feeling.

Experience

Combat Designer

Unreal Engine C++ Programmer

Houston, TX

June. 2023 – Present

- Created a custom movement system in Unreal Engine that blends scripted movement, animation root motion, and raw physics, delivering responsive, fluid character control that supports input buffering, aerial combat, and enemy manipulation for combos and directional launches.
- Developed a modular targeting system in Unreal Engine, allowing abilities and gameplay events to dynamically adjust weights and parameters, enabling intelligent target selection, adaptive camera perspectives and angles, and cinematic shots.
- Engineered an attack feedback system using hit reactions, visual effects, and sound cues to ensure every strike communicates impact and weight, with customization such as per-attack slash ribbons and audio cues to reward hit confirmation and combo payoff.

British Petroleum (BP)

Data Engineer

Houston, TX

June. 2024 – Oct. 2025

- Engineered event-triggered data ingestion pipelines in Azure Data Factory to process raw Navisworks files into structured metadata via Databricks, integrating the results into BP's Enterprise Knowledge Graph and powering Virtual Asset Management, a near 1:1 interactive digital twin of on-site rigs to deliver contextualized data across the enterprise.
- Assisted in a DevOps team for a Data Modeling and Contextualization team, implementing CI/CD pipelines and integrating tools like Azure DevOps, JFrog, and Checkmarx to accelerate deployment cycles from development to production while ensuring security and code quality at scale.
- Collaborated cross-functionally across many teams, such as Product Management and Optimization, enabling data ingestion through Azure Data Factory and Databricks while extending DMCDT-developed services, including event-triggered pipelines to unify workflows across BP's digital ecosystem.

Education

University of Houston

B.S. Computer Science

Houston, TX

Aug. 2020 - Graduated

Notable Coursework:

Artificial Intelligence, Linear Algebra, Software Design, Database Systems, Operating Systems, Computer Networks, Data Structures and Algorithms, Numerical Analysis

Wharton County Junior College

A.S. Computer Science

Sugar Land, TX

Aug. 2018 - Aug. 2020

Skills

Technologies: Unreal Engine 4/5, Perforce, Git, GitHub, Databricks, Azure Dev Ops, Azure Data Factory, VMs, CI/CD, Linux, Agile, REST APIs

Languages: C++, C, Python

Databases: MySQL, Microsoft SQL Server, SQL Alchemy, MongoDB